The Short, the Long, and the Consequences of VAR Analysis For Structural DSGE Estimation

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Motivation

- VARs can be used to estimate DSGE frameworks by IRFs minimization matching
- If the VAR is mispecified the DSGE might also be so

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  If the ***VAR is mispecified****, however, then the* ***specification errors will be computed at each horizon****"* **Ramey2016), p. 83**
- VARs and DSGEs are instruments used by central banks to make policy decisions
- Misspecification can deliver distorted results mis informing the central banker

This Paper

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Findings:

- Both VAR based IRFs and structural DSGE parameter estimates are significantly affected by the inclusion of longer lags

VAR IRFs DSGE IRFs DSGE Parameters

Related Literature

Long-Lag VARs:

- (De Graeve and Westermark2025; Baumeister2025; Antolín-Díaz and Surico2025; Antolín-Díaz et al.2026)

Truncation Bias:

- (Ravenna2007; De Graeve and Westermark2025; Chari et al.2008; Barnichon et al.2026)

DSGE State Space representation into VAR(∞)

- (Fernández-Villaverde et al.2005; Ravenna2007; Giacomini2013; Theodoridis2011; Forni et al.2016)

IRFs:

- (Ludwig2026; Baumeister2025; Olea et al.2025; Plagborg-Møller and Wolf2021; Li et al.2024; Òscar Jordà2025)

DSGE IRFs matching with VARs

- (Christiano et al.2005; Altig et al.2011; Rossi et al.2012)

Illustration of the problem: Truncation Bias

$$Y_t = B_1 Y_{t-1} + \cdots + B_p Y_{t-p} + u_t, \qquad \mathbb{E}(u_t u_t') = \Sigma \tag{1}$$

$$B(L) = B_1 L + \cdots + B_p L^p \tag{2}$$

- The VAR with $p < p^*$ suffers from truncation bias.[1]
- Lag truncation biases the reduced form coefficients $B(L)$ and the reduced form variance covariance matrix Σ, which are the two main ingredients for IRF estimation.

$$\text{IRF} = f\big(B(L), \Sigma\big) \tag{3}$$

[1] Where p^* denotes the DGP true lag length.

DSGE, State Space and VAR(∞) Representation

- Many DSGE's log linearized solution can be mapped in State Space representation and then into VAR(∞)
- If a DSGE is estimated via IRFME with a finite order VAR, the DSGE is estimate with a Biased by Truncation VAR

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- **Should the VAR's lag order increased?**

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Lower Variance

- Misspecification reduction lowers the variance of the residuals
- This variance reduction counteracts the imprecision caused by increased parametrisation

Long-Lag

Missing file: InSample_Residuals_Fed_Funds_Rate.pdf

Figure 1: FFR Residuals Comparison Across lag 4 Vs 12

Long-Lag Benefits

The variance-reducing effect of lower misspecification often dominates the penalty of increased parameterisation.

Compared to short-lag VARs, Long-Lag VARs have:

- Reduced truncation bias
- More precise inference
- Lower MSE
- Better coverage

Replication Exercise

Framework:

- **Altig et al.2011)**'s DSGE[2]

VAR:

- Medium scale VAR: 10 variables
- Monetary Policy, Neutral and Embodied Technology shocks
- Sample period 1959:Q2-2001:Q4

Exercise:

- I re-estimate the VAR increasing lag order
- Re-estimate the DSGE by IRFs matching
- **Analyse implications of possible variations in the structural parameters and IRFs across different lag choice**

[2]Where the replication package provided by the authors refers to the 2005 NBER working paper

ACEL's DSGE Mechanism

Classic NK models account for inflation inertia by assuming that firms re-optimise their prices every six quarters or even less often

- In Altig et al.2005) inflation is inertial despite firms re-optimising prices on average every 1.5 quarters
- Inertia is explained by the fact that, when firms re-optimise, they do it by a small amount because each firm's short run marginal cost is increasing in its own output
- The key assumption is that, in any given period, a firm's capital stock is pre-determined and firm-specific

They build two models on Christiano et al.2005):

- Homogeneous capital
- Firm-specific capital

ACEL's DSGE Mechanism (Cont'd)

- When log-linearized, the only difference in the characterizing equilibrium equations lies in the Phillips Curve

$$\Delta\hat{\pi}_t = \mathbb{E}[\beta\Delta\hat{\pi}_{t+1} + \gamma\hat{s}_t|\Omega_t] \tag{4}$$

$$\gamma = \frac{(1-\xi_p)(1-\beta\xi_p)}{\xi_p}\chi \tag{5}$$

- $\hat{s}_t$ is the economy-wide marginal cost, γ the Phillips Curve slope, ξ_p the Calvo price stickiness parameter
- χ reduces the sensitivity of inflation to marginal costs in the firm-specific capital model
- The two models are observationally equivalent in terms of prices and quantities: macro data identify γ, so the difference between the two models lies in the implied ξ_p

ACEL's DSGE Mechanism (Cont'd)

- In the **Homogeneous** model, $\chi = 1$: γ depends only on β and ξ_p
- In the **Firm-specific** model, $\chi < 1$ and captures two forces that make a firm move its price *less* when it can reset:
  - the costlier it is to vary capital utilization, the steeper the firm's own marginal cost curve
  - the more elastic the firm's demand, the larger the fall in output (and hence in marginal cost) after a price increase

- So χ governs **how much** a firm moves its price
- ξ_p **how often**, with implied duration $1/(1 - \xi_p)$

Since γ is estimated by aggregate data, a smaller χ must be imply a smaller ξ_p

ACEL(2005) Results

The estimated γ is 0.040, which implies:

- **Homogeneous** capital: $\xi_p = 0.82$, firms change prices every 5.6 quarters
- **Firm-specific** capital: firms change prices every 1.5 quarters

The gap arises because firms' demand curves are highly elastic and their marginal cost curves are very steep: the same γ is delivered by frequent but small price adjustments.

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- **Are these results robust to an enriched lag structure of the VAR?**

Thanks a lot for your attention!

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VAR IRFs; Monetary Policy Shock

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Figure 2: IRFs VARs all CI Comparison: Monetary Policy Shock